

Control of active power fed into the grid

The derivative of the angle θ is the frequency ω .

Consider a steady state condition and a grid voltage constant in amplitude and frequency: $\overline{v_g} = V_g e^{j\omega t}$. Looking from the reference frame (dq), all the quantities are constant. Therefore:

$$\dot{\theta} = \omega$$

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

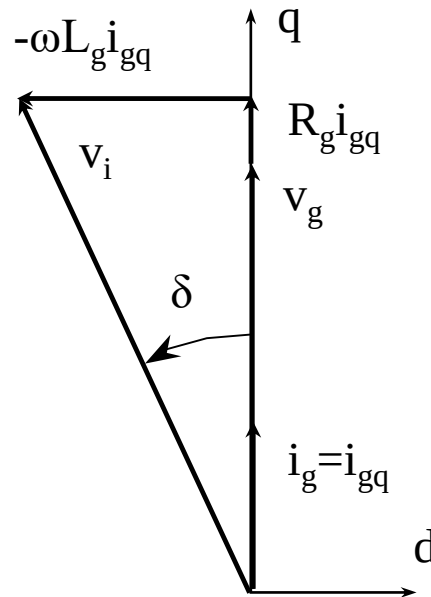
$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

If the current i_{gd} (component along the d axis) is fixed to zero, the phasor diagram is:

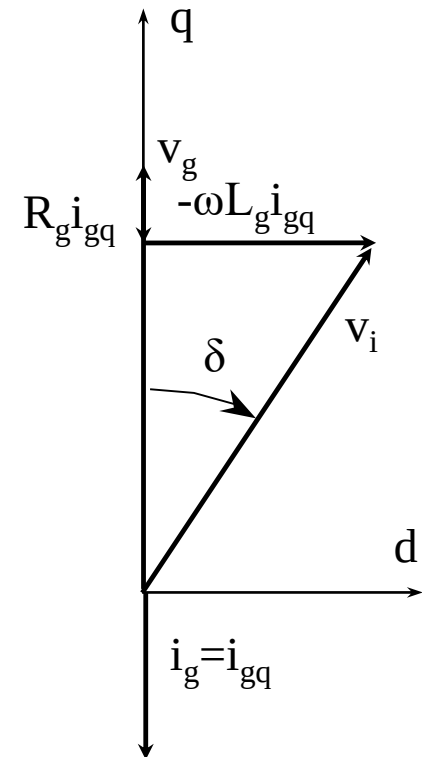
$$v_{id} = -\omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + V_g$$

Regenerative & $Q_g=0$



Rectifier & $Q_g=0$



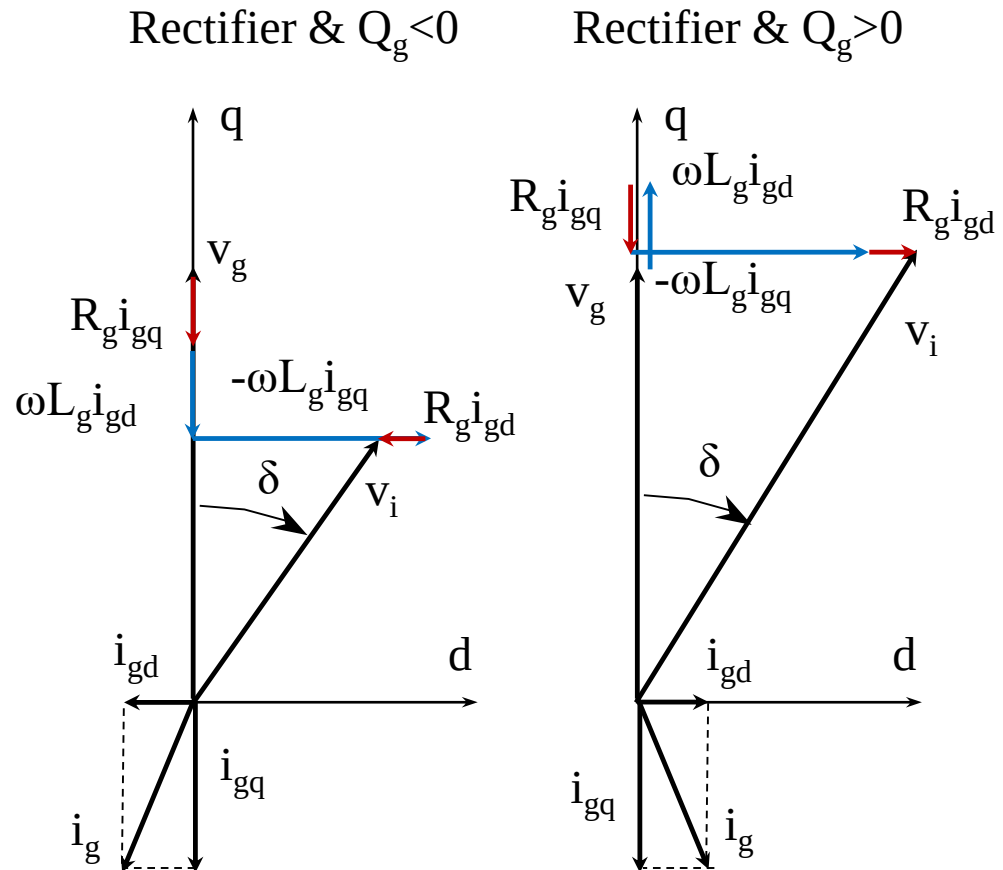
Control of active/reactive power fed into the grid

Controlling the current i_{gd} , it is possible to drive the reactive power: a positive value means that the inverter looks like a capacitor (and the grid like an inductor); vice versa for a negative value.

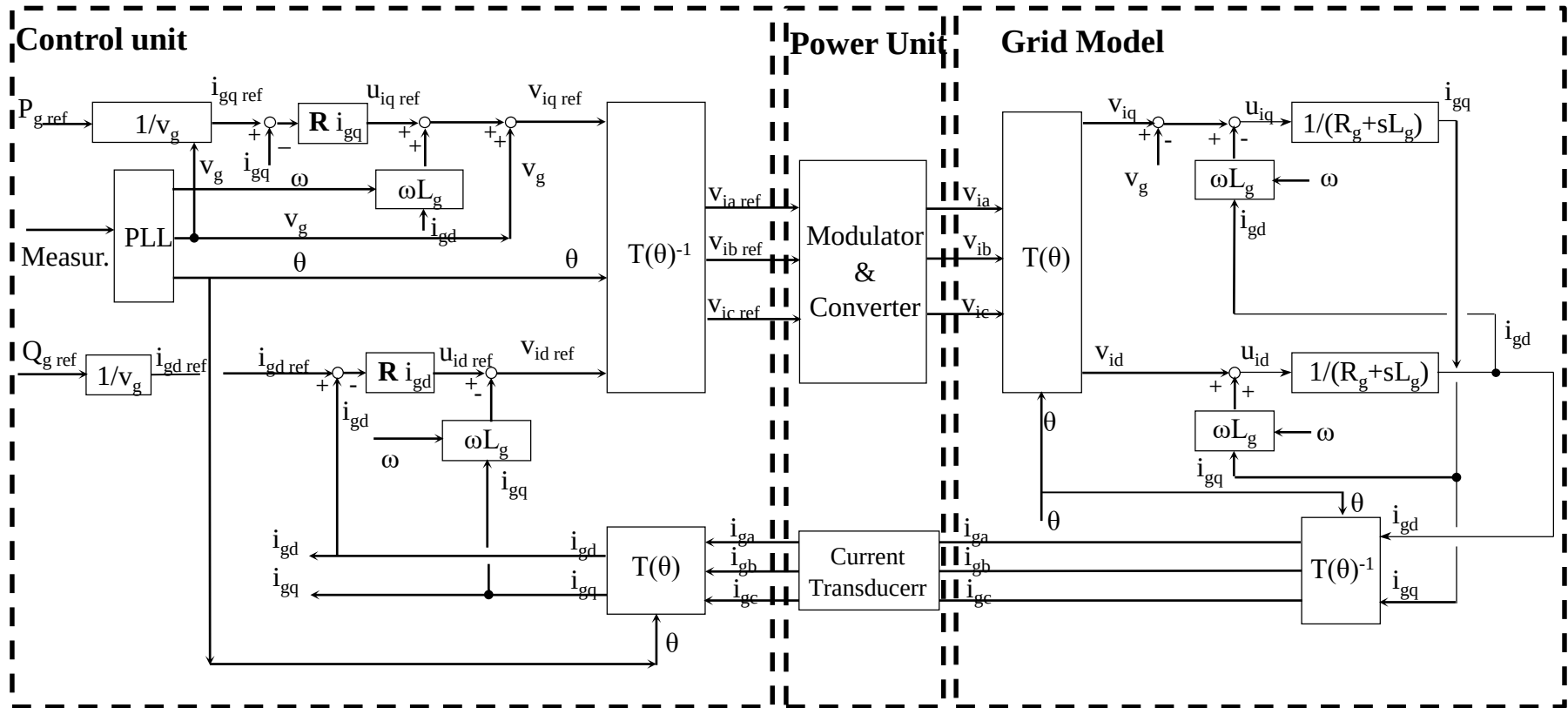
$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

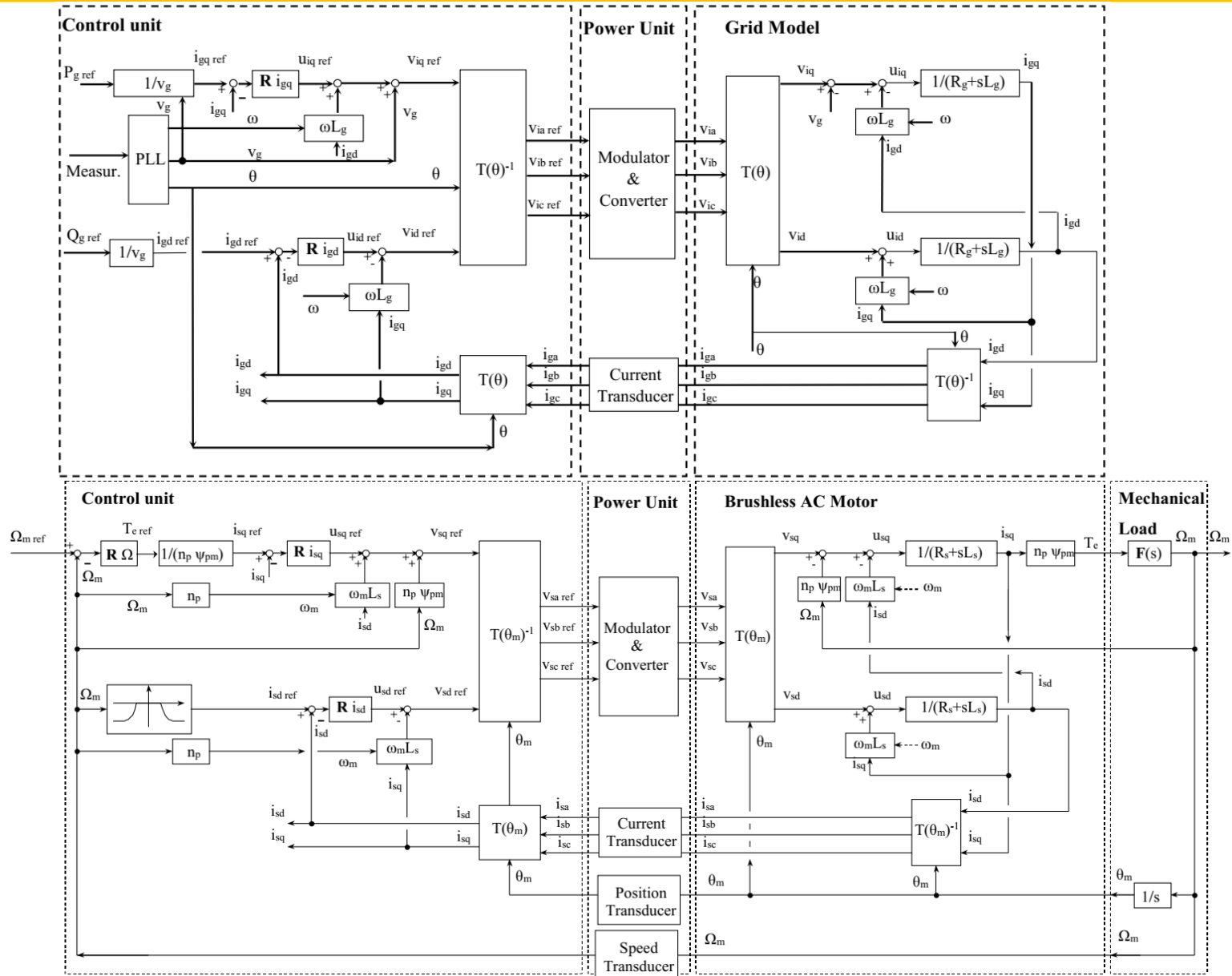
The figure represents the conditions when the active power is negative (operation as rectifier). Similar approach may be done for positive value of the active power (regeneration)



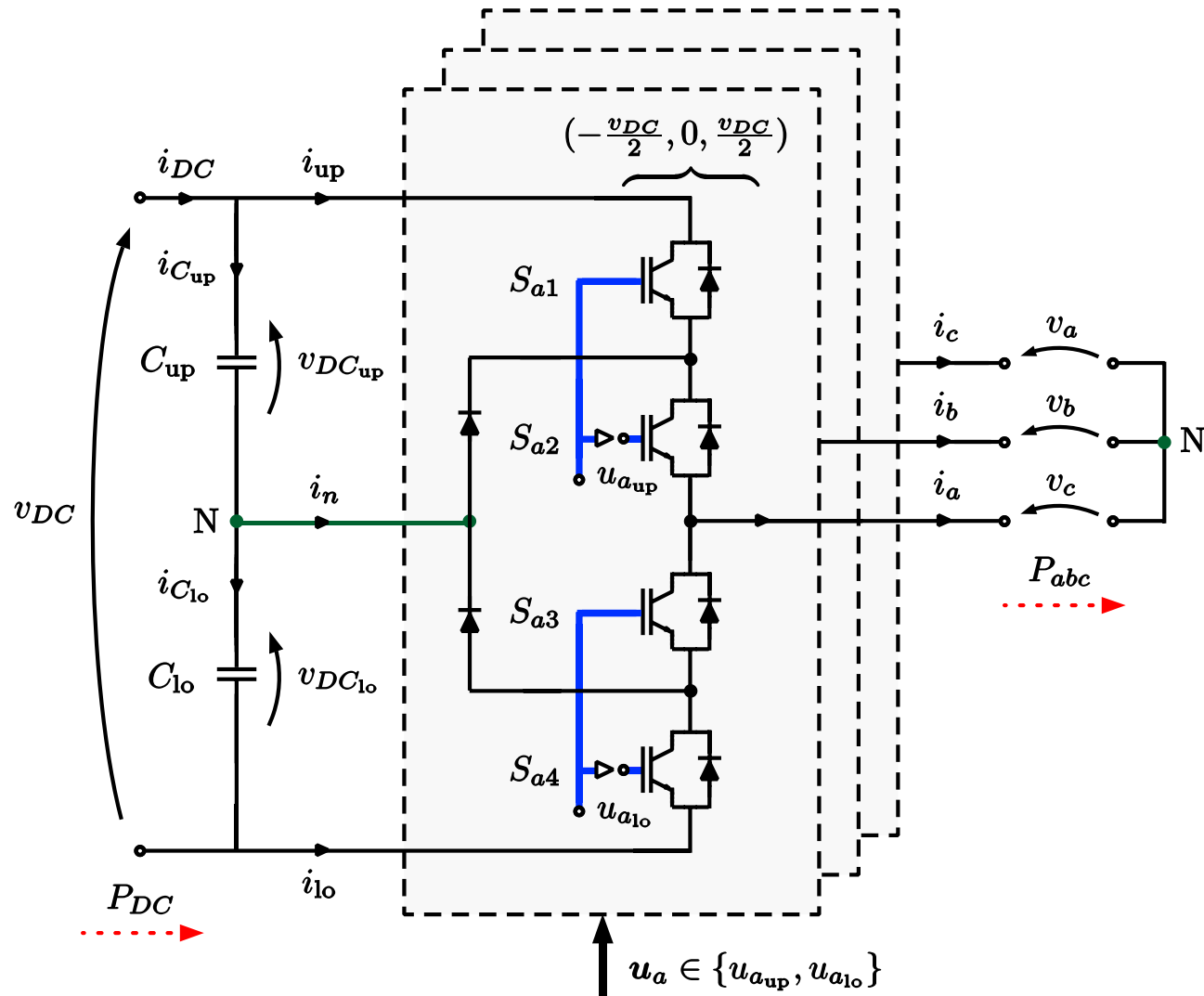
Control schemes



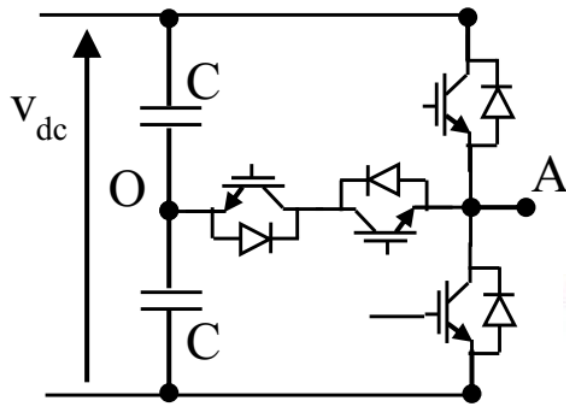
Control schemes: comparison with ac brushless



Multilevel converter: 3L-NPC

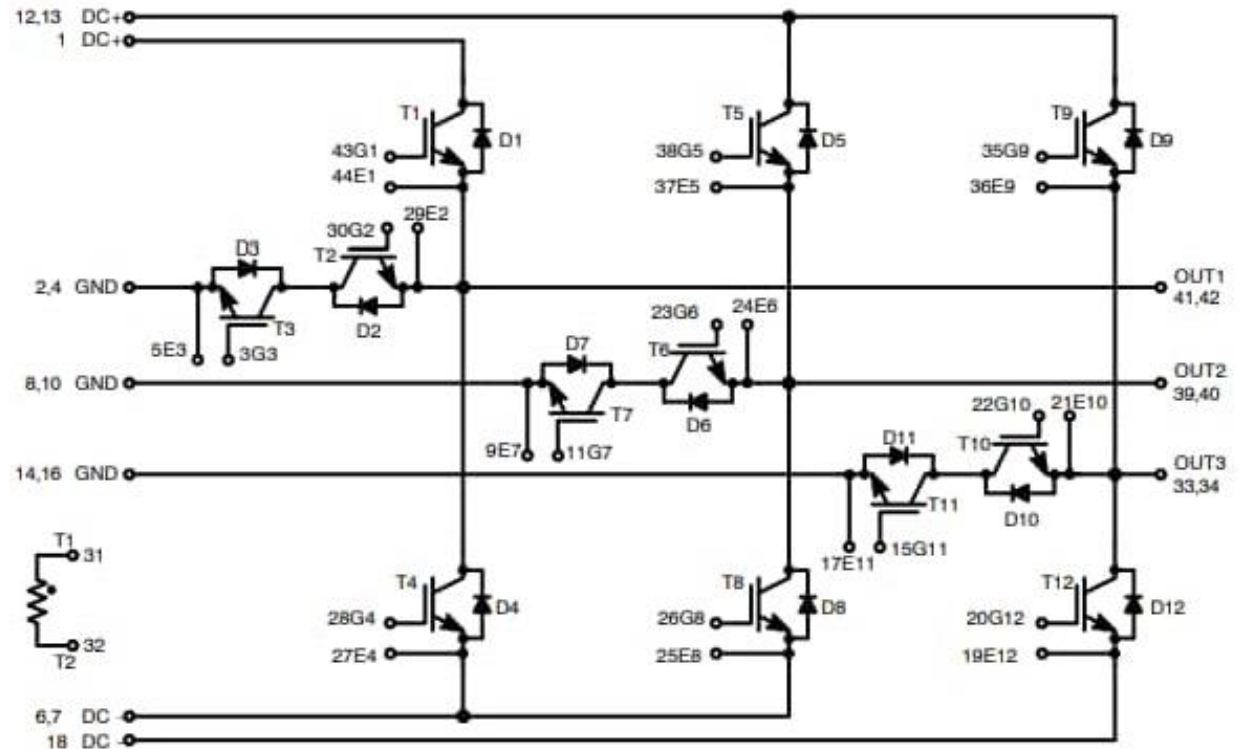


Multilevel converter: 3L-TNPCC



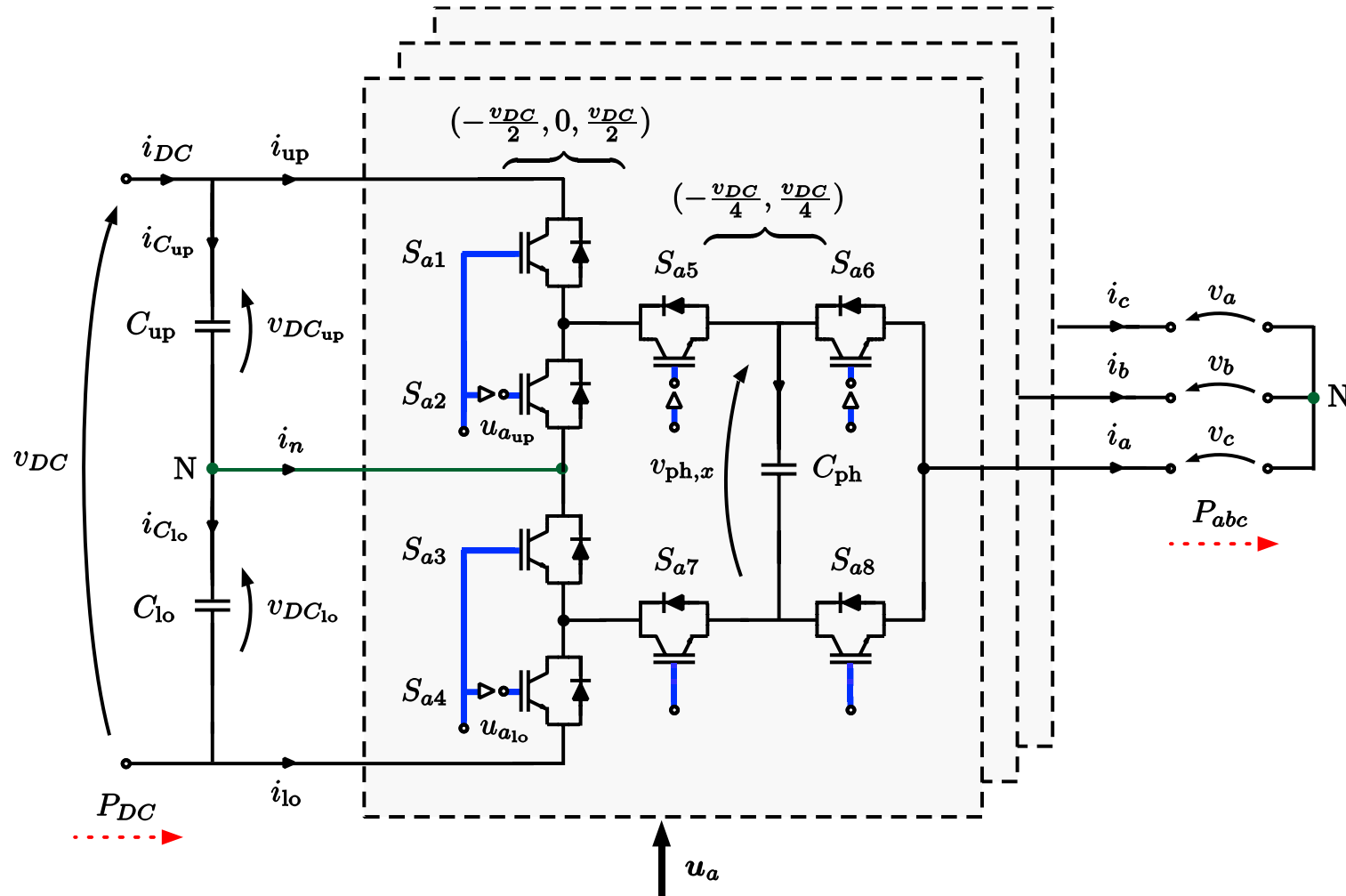
One leg

Three legs

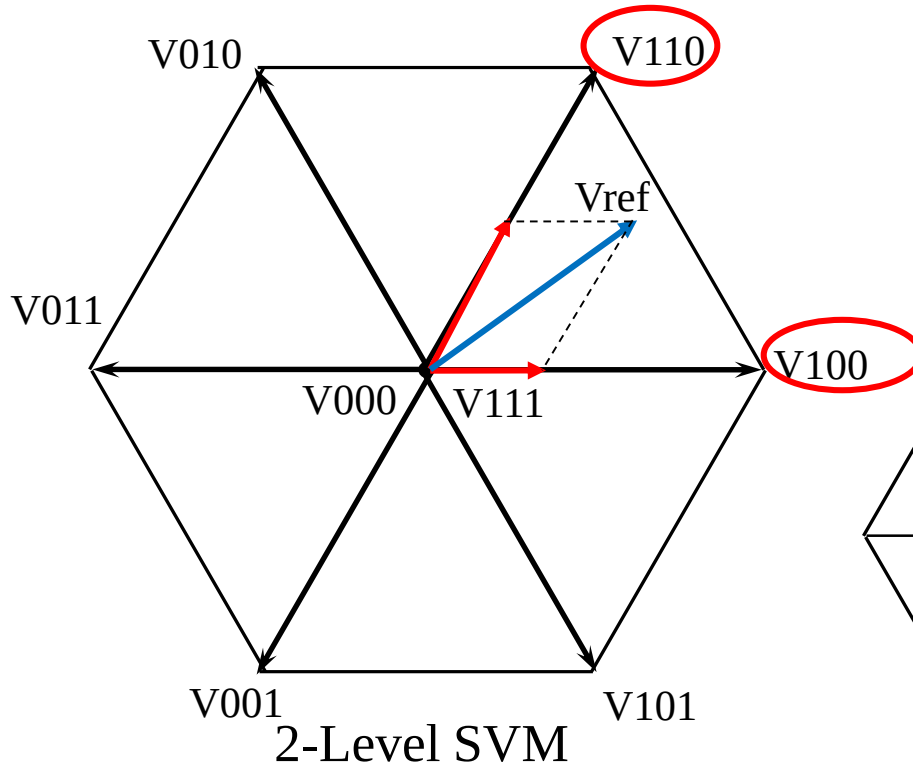


ON Semiconductor NXH40T120L3Q1 3-Phase TNPC Module

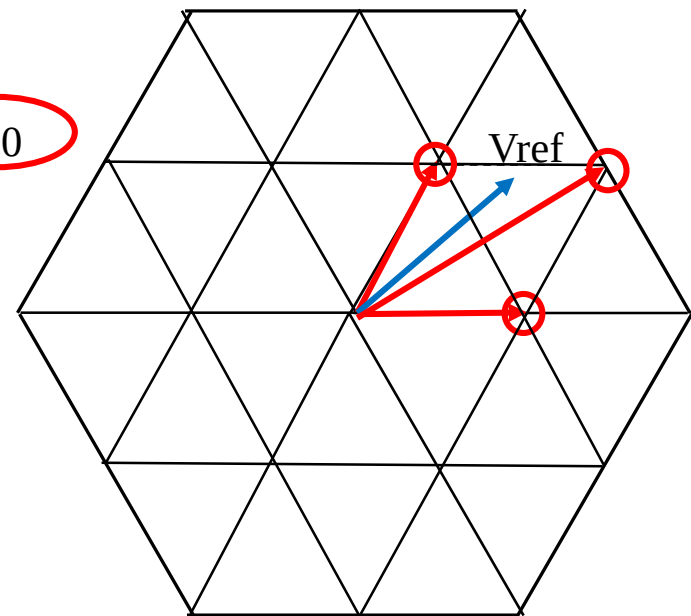
Multilevel converter: 5L-NPC



SVM for a 2L-VSC and 3L-NPC



3-Level SVM



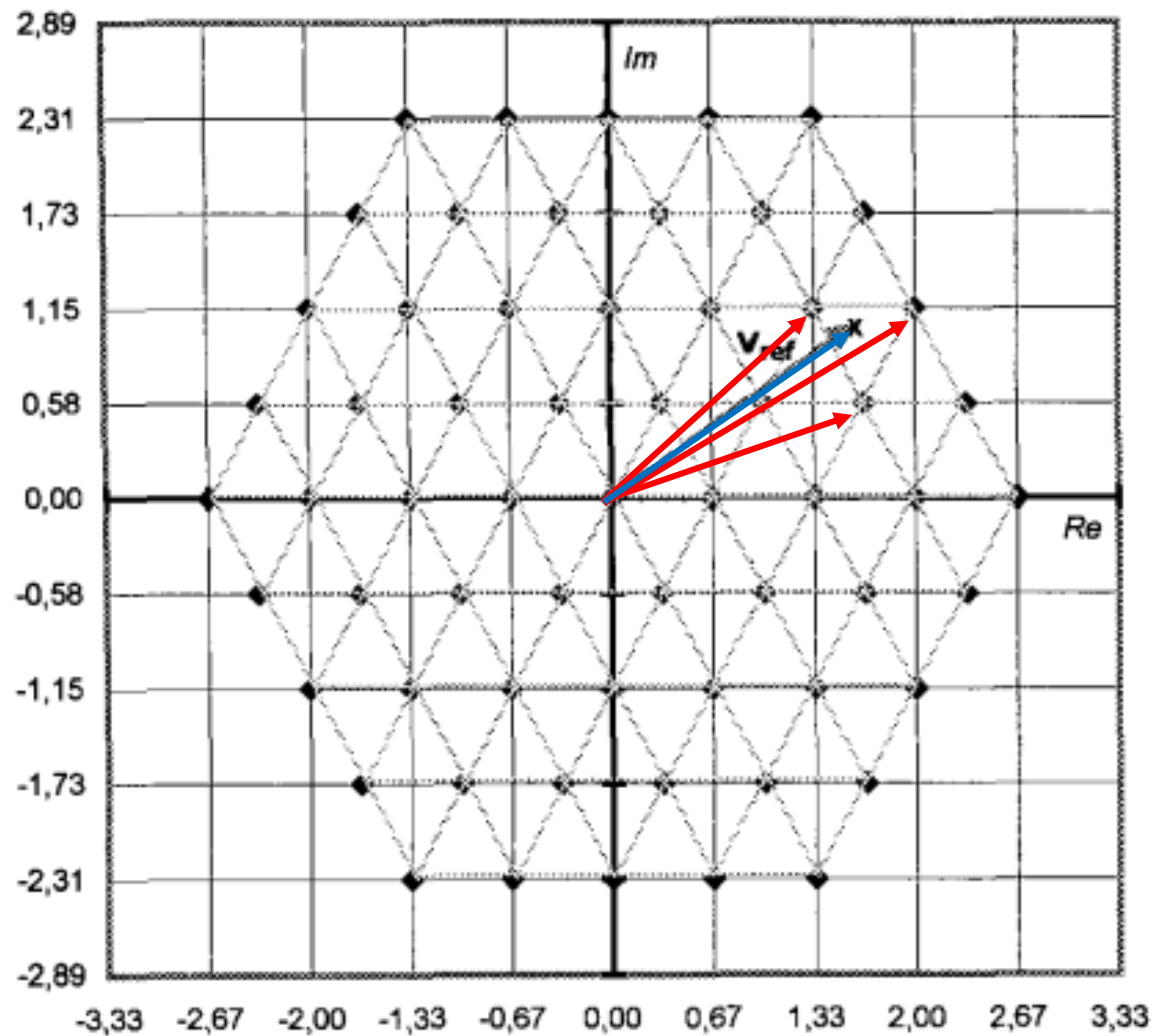
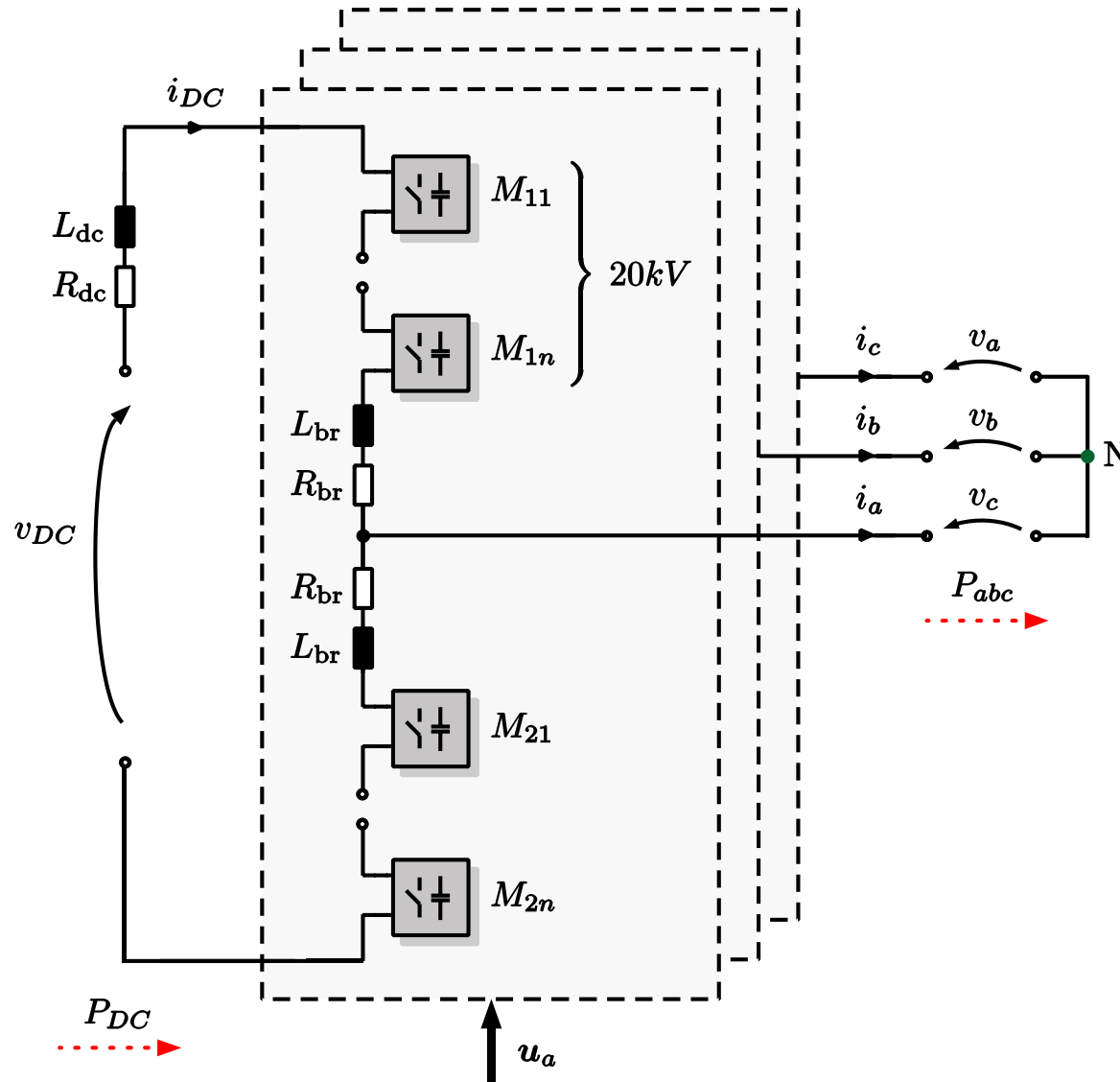
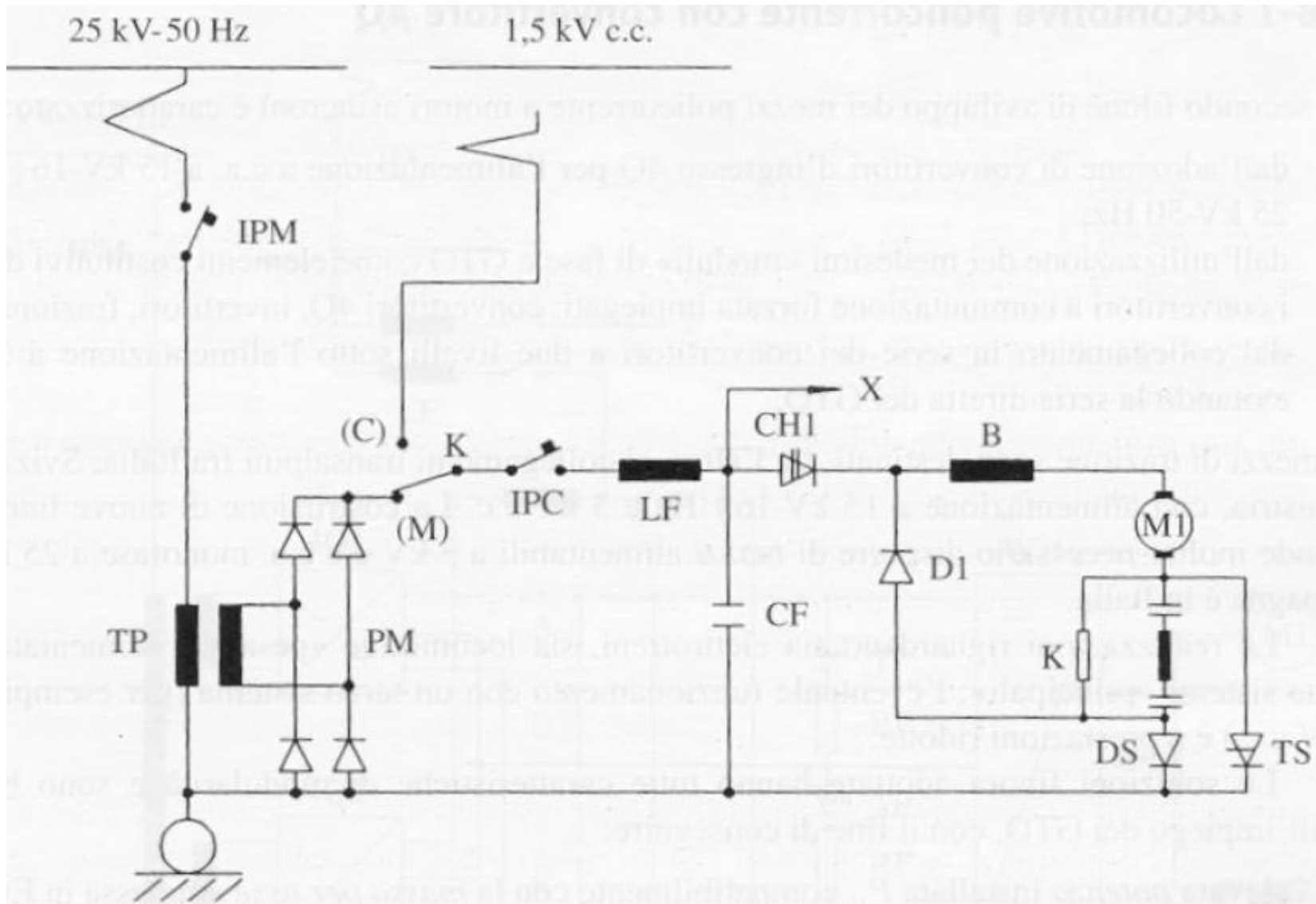


Fig. 3 Space-vector diagram of a 5-level converter

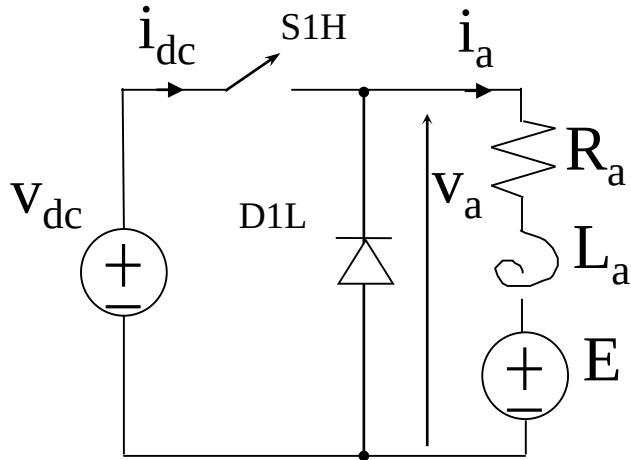
Multilevel converter: MMC



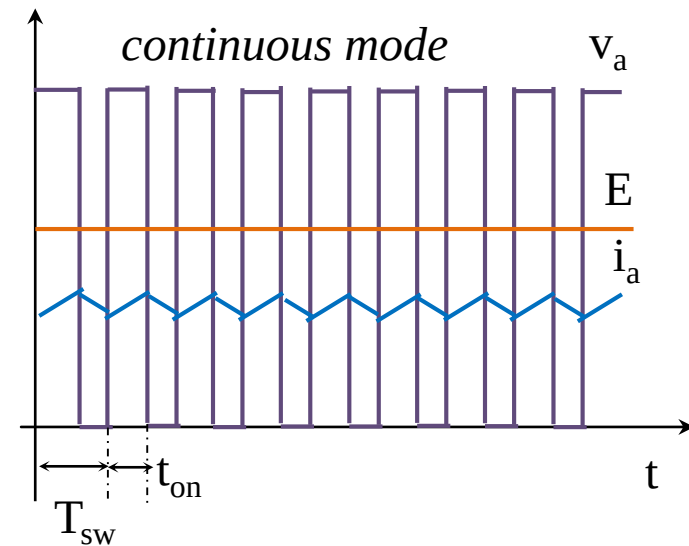
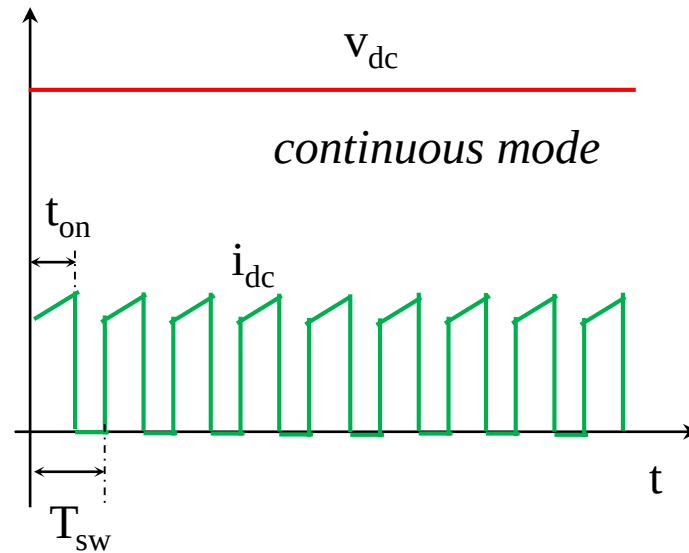
DC/DC: Chopper

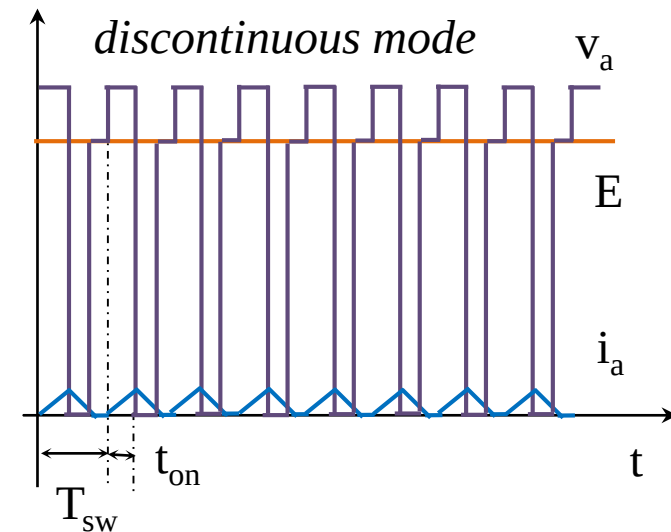
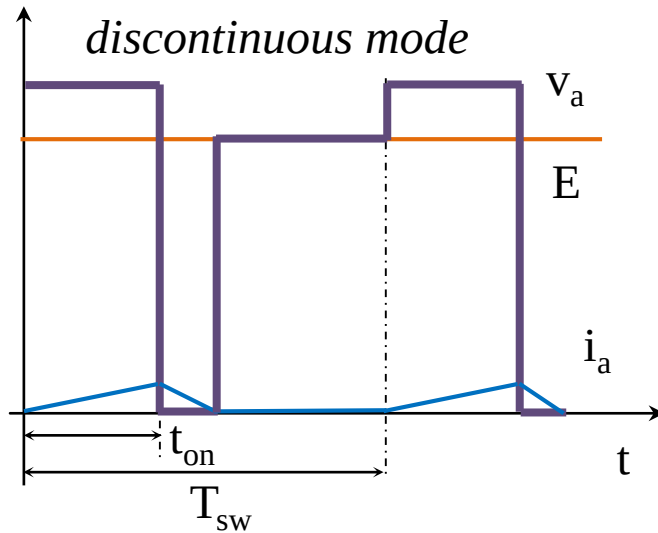
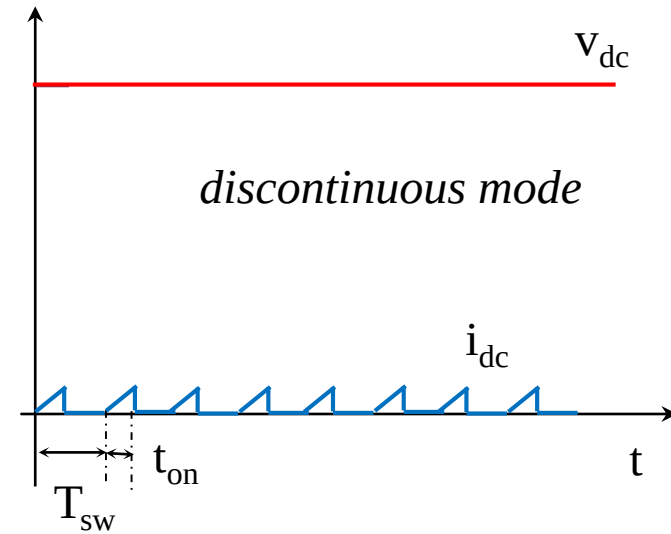
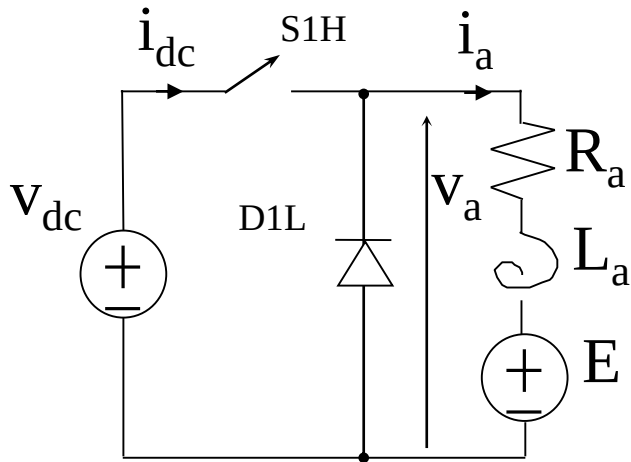


CH1: static switch, D1 free-wheeling diode



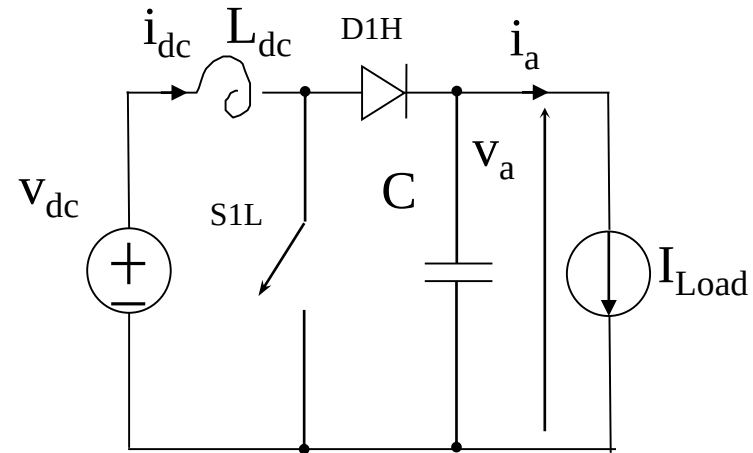
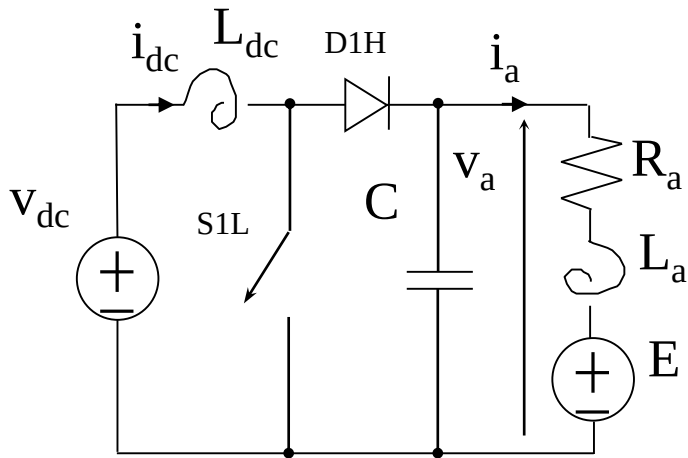
- Buck
- Step-down
- Chopper
- 1-quadrant



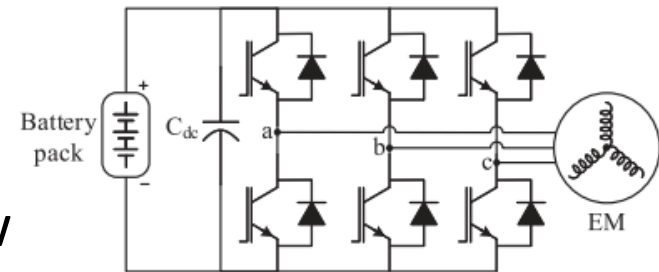


The output voltage is **not proportional** to the duty cycle

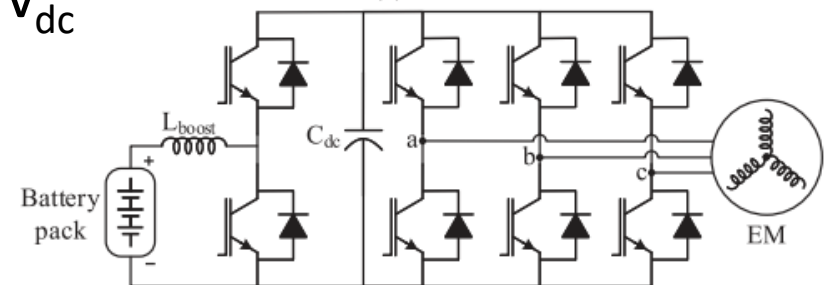
DC/DC: boost converter



- Also called step-up
- Used when the battery voltage is low
- The output voltage is higher than v_{dc}
- A soft-starter is needed
- PFC has similar characteristics
- Operating region

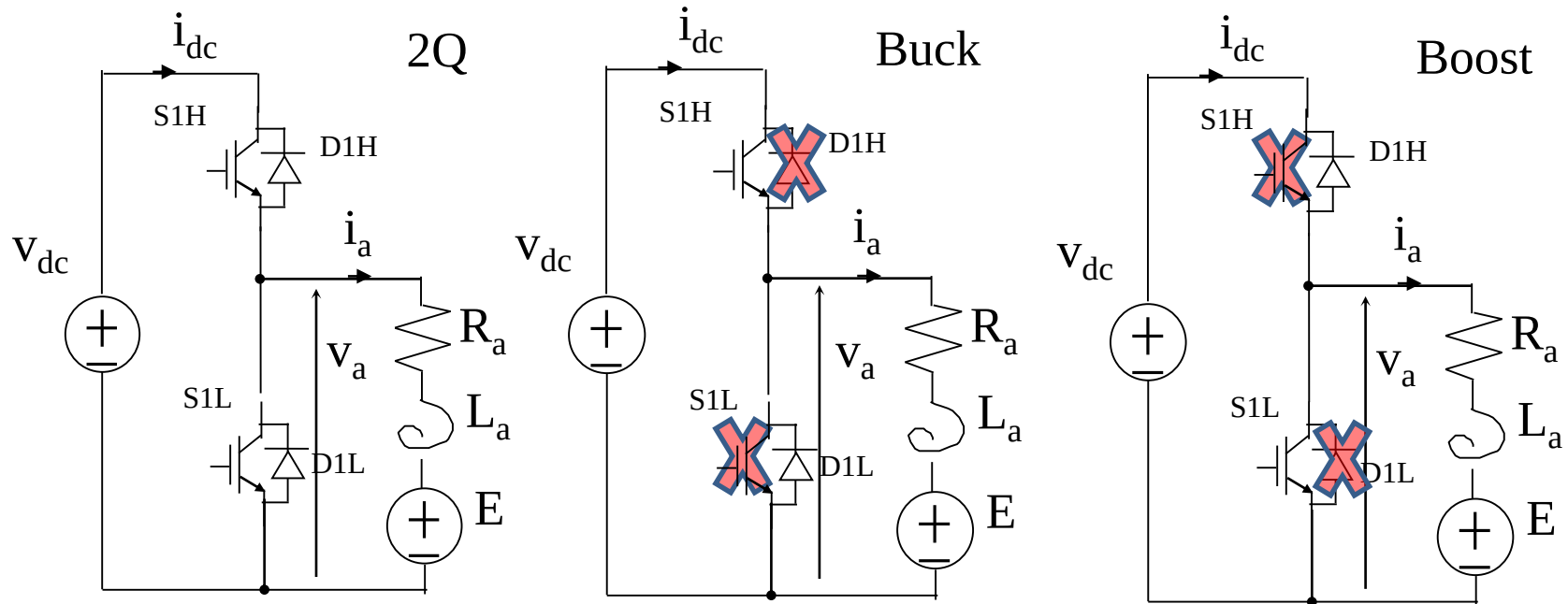


(a) VSI.



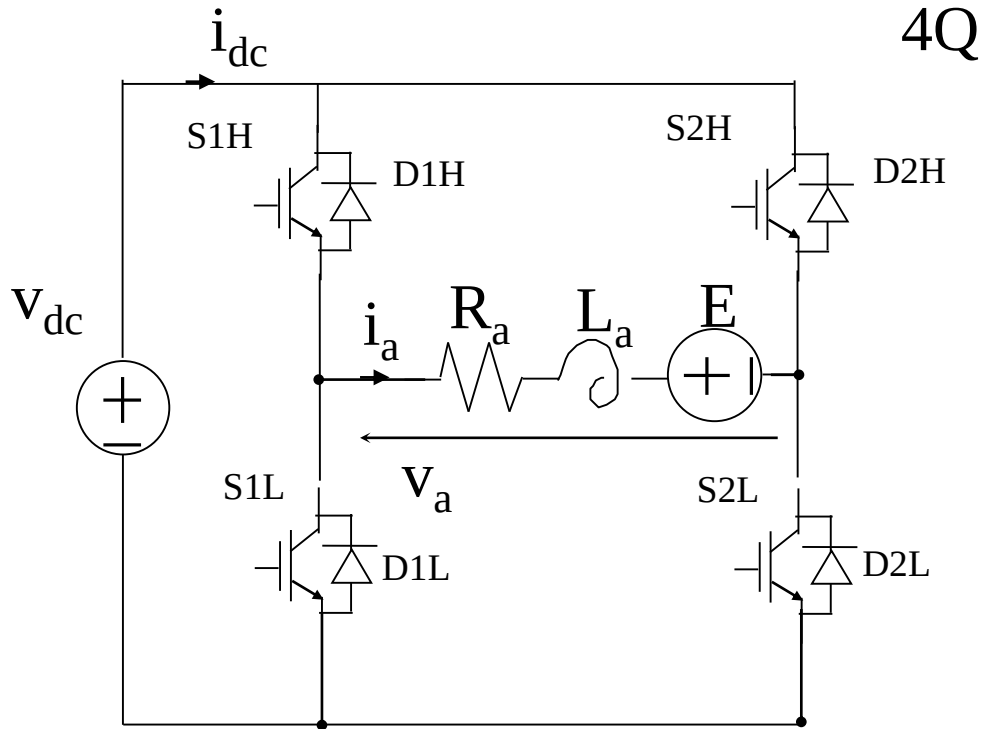
(b) VSI combined with a DC/DC boost converter.

DC/DC: half bridge



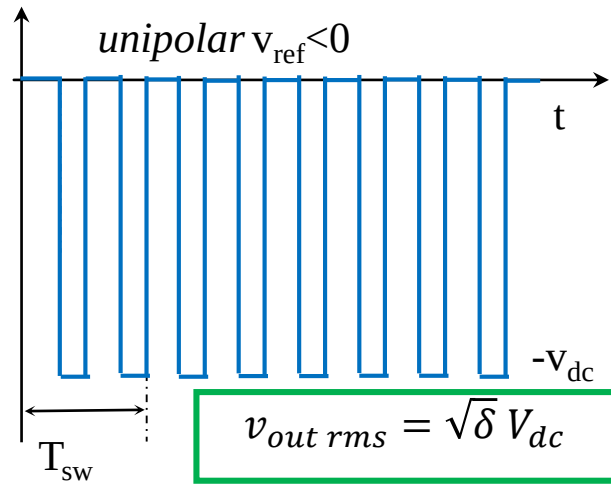
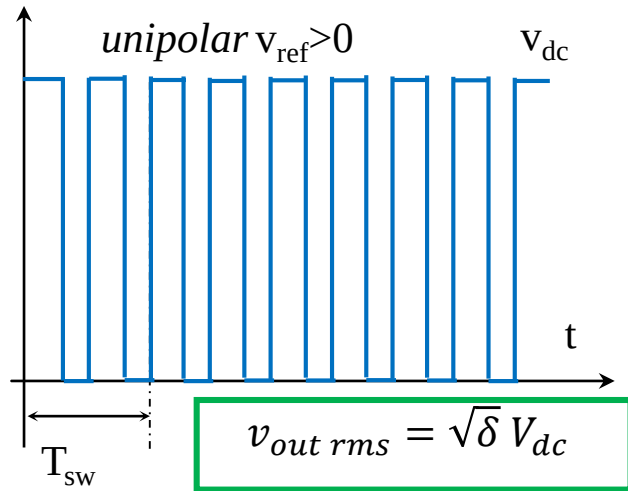
- Two quadrants converter: v_a always positive, i_a positive and negative
- From the mechanical point of view: speed always positive, torque positive (motor) or negative (brake)

DC/DC: Full bridge (H-bridge)

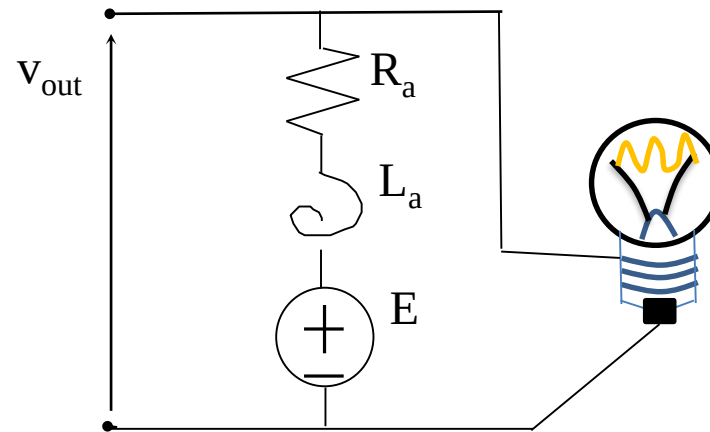
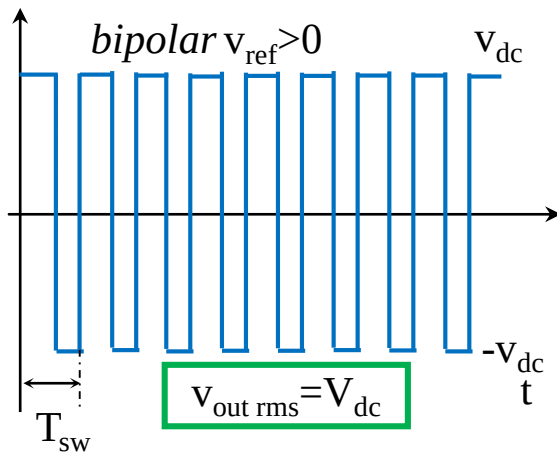


- Four quadrants converter (4Q): v_a always positive and negative, i_a positive and negative
- From the mechanical point of view: speed positive and negative, torque positive (motor) or negative (brake)

Modulation strategies: unipolar or bipolar



$$\Delta I_{uni} \cong \frac{1}{4} \Delta I_{bip}$$



Isolated DC/DC converters

- Recharging device (station or on-board system)
- Auxiliary supply (trains)

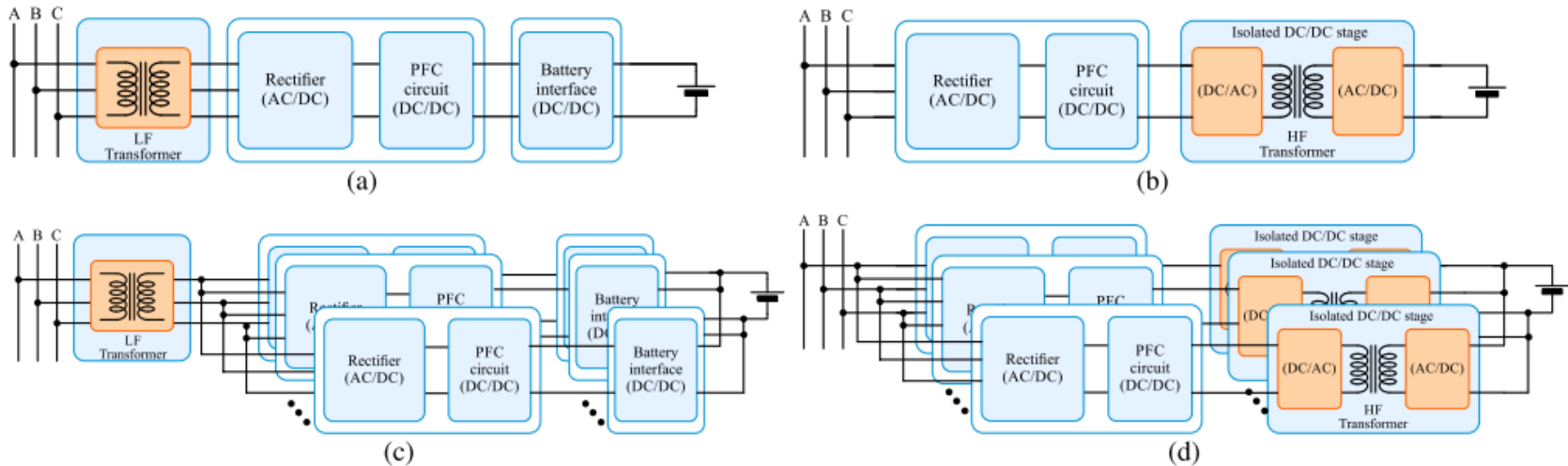
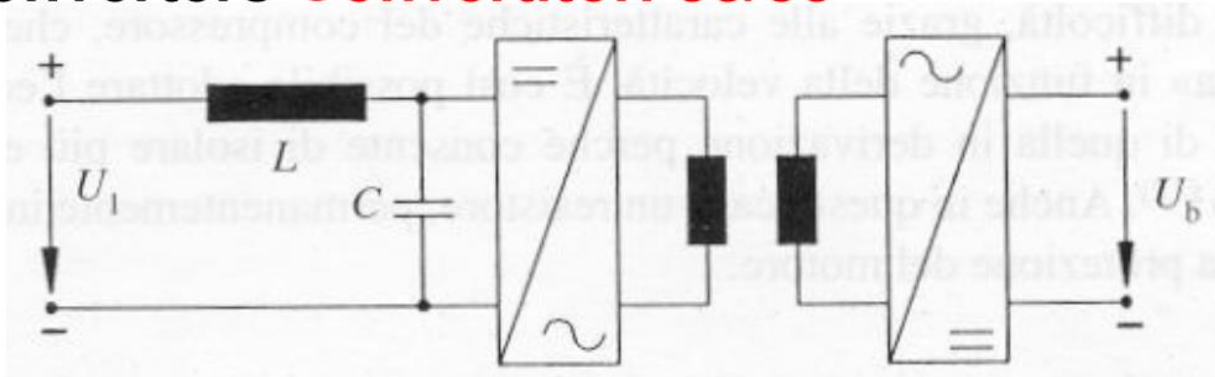
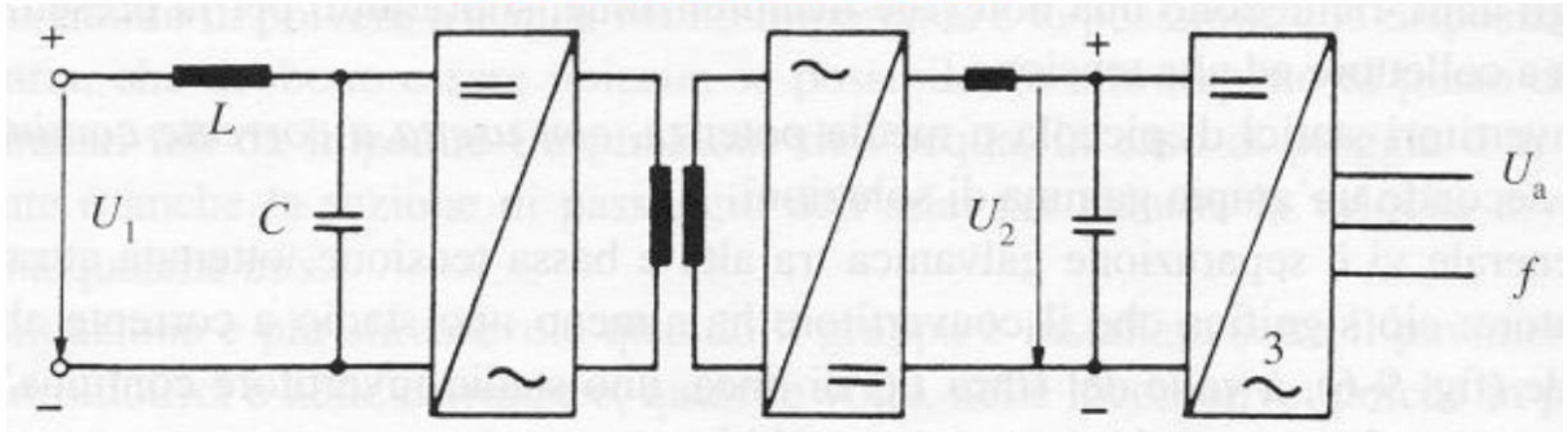


Fig. 1. Simplified block diagram of conventional dc fast charger power conversion systems. (a) Single-module charger with a nonisolated dc/dc converter. (b) Single-module charger with an isolated dc/dc converter. (c) Multiple paralleled modules shown in (a). (d) Multiple paralleled modules shown in (b).

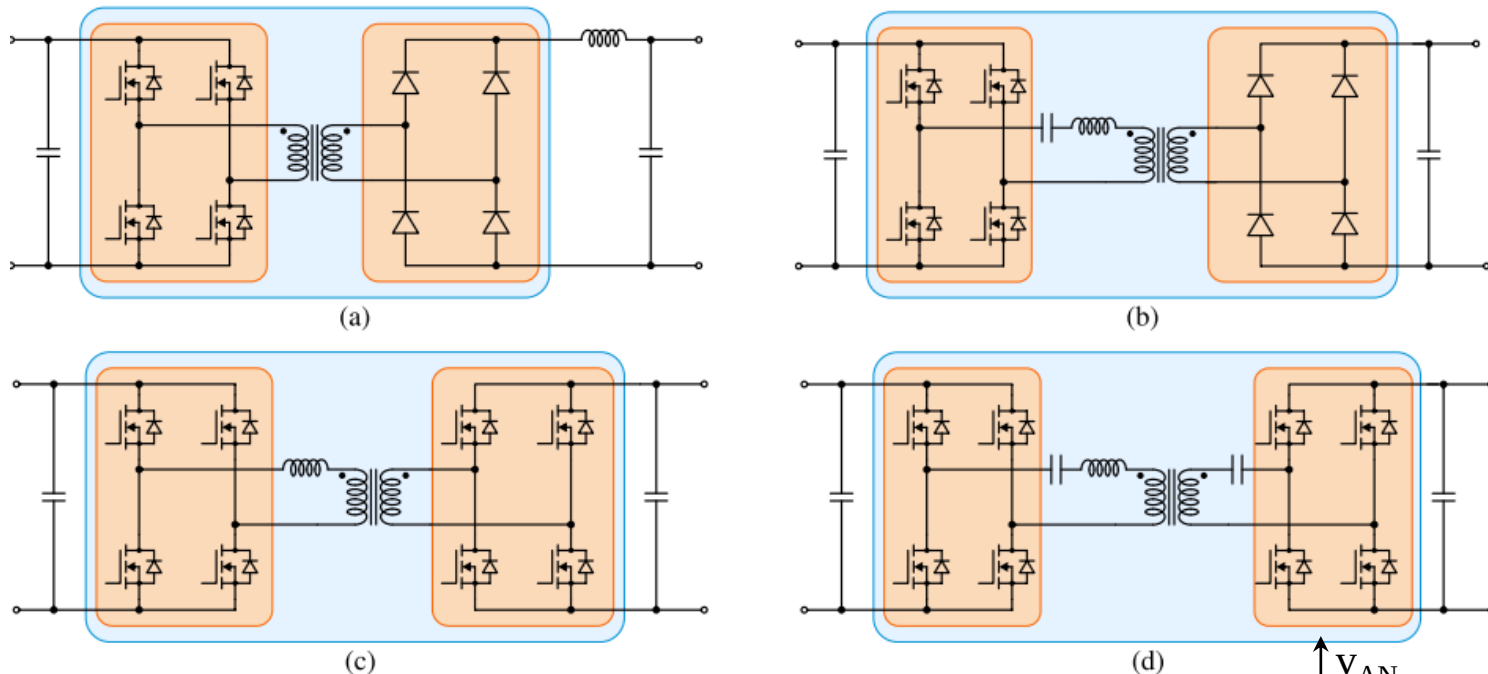
DC/DC converters **Convertitori cc/cc**



DC/AC converters **Convertitori cc/ca**

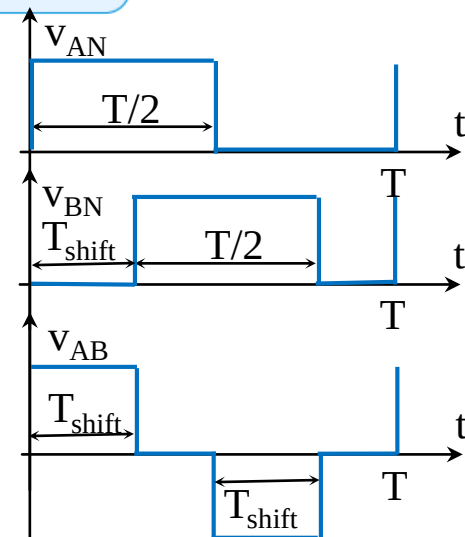


SST: solid-state transformer

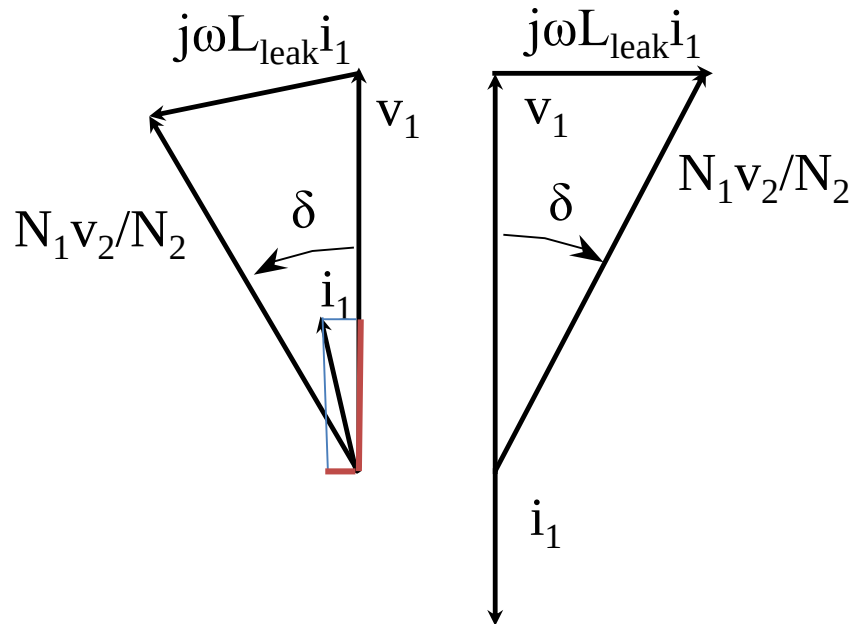
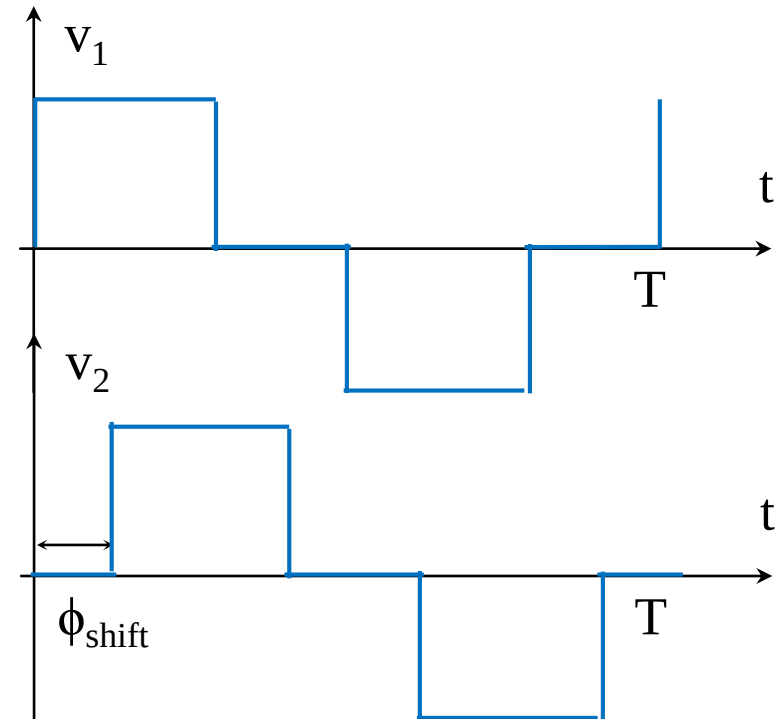
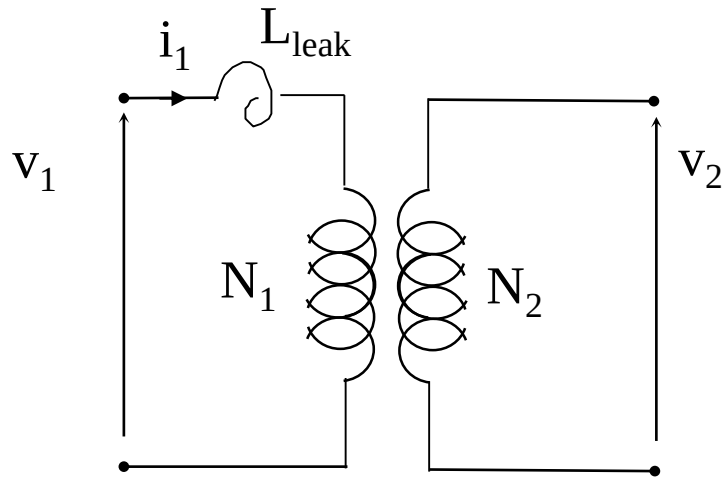


Isolated dc-dc converter topologies for dc fast chargers

- Phase-shift full-bridge (PSFB)
- LLC resonant converter
- Dual active bridge (DAB)
- CLLC resonant converter



Dual active bridge control technique



High power configuration

